Activity sheet 1.4 Rules for balancing equations and writing formulae for ionic compounds

Rules for balancing equations

1. Write a word equation for the reaction.
2. Write the formula for each substance using the rules for covalent or ionic substances.
3. Balance the equation by putting numbers in front of formulae so that the numbers of atoms of each element is the same on reactant and product sides.

   Do this in the following order:

   (i) Balance metal atoms

   (ii) Balance non-metal atoms

   (iii) Balance hydrogen atoms

   (iv) Balance oxygen atoms

   (v) Add subscripts for states (s), (l), (g) or (aq).

1. Balance the following equations by adding numbers in front of formulae:

a. $Mg_{(s)} + H_2O_{(i)} \rightarrow Mg(OH)_{2(aq)} + H_{2(g)}$

b. $H_2S_{(g)} + H_2SO_{4(aq)} \rightarrow S_{(s)} + SO_{2(g)} + H_2O_{(l)}$

c. $FeCl_{2(aq)} + Cl_{2(g)} \rightarrow FeCl_{3(aq)}$

d. $Fe_{(s)} + HNO_{3(aq)} \rightarrow Fe(NO_3)_{2(aq)} + H_{2(g)}$

e. $(NH_4)_2SO_{4(aq)} + BaCl_{2(aq)} \rightarrow NH_4Cl_{(aq)} + BaSO_{4(s)}$

f. $CH_{4(g)} + O_{2(g)} \rightarrow CO_{2(g)} + H_2O_{(l)}$

g. $Al_{(s)} + Cl_{2(g)} \rightarrow AlCl_{3(s)}$

h. $Zn_{(s)} + HNO_{3(aq)} \rightarrow Zn(NO_3)_{2(aq)} + H_{2(g)}$

2. Write balanced equations for the following reactions.

   a. calcium + oxygen → solid calcium oxide

   b. hydrogen chloride gas + sodium hydroxide solution → sodium chloride solution + water

c. sodium + water → sodium hydroxide + hydrogen

d. magnesium carbonate powder → magnesium oxide powder + carbon dioxide

e. sodium + oxygen → solid sodium oxide

f. zinc + copper(II)sulfate solution → zinc(II)sulfate solution + copper

g. zinc + hydrochloric acid → zinc(II)chloride solution + hydrogen

3. Write equations to represent the following reactions.

a. Dilute hydrochloric acid and sodium hydroxide solution react to produce a solution of sodium chloride and water.

b. Burning carbon in a limited amount of oxygen produces carbon monoxide gas.

c. Dilute hydrochloric acid and sodium carbonate solution react to produce a solution of sodium chloride, carbon dioxide and water.

d. Sodium hydroxide solution is added to dilute sulfuric acid to produce a solution of sodium sulfate and water.

e. Solutions of potassium iodide and lead(II)nitrate are mixed to form a precipitate of lead iodide and potassium nitrate solution.

f. Copper metal is added to hydrochloric acid to produce copper(II)chloride solution and hydrogen.

g. Sulfur dioxide gas reacts with oxygen to produce sulfur trioxide gas.

h. Solid calcium carbonate decomposes when heated to form solid calcium oxide and carbon dioxide gas.

i. Nitrogen monoxide reacts with oxygen to produce nitrogen dioxide gas.

j. Calcium bromide and ammonium carbonate solutions react to produce a precipitate of calcium carbonate and ammonium bromide solution.

k. Dinitrogen tetroxide gas decomposes to form nitrogen dioxide gas.

Rules for writing formulae for ionic compounds.

1. Check that the compound is ionic (made from a metal and non-metal ion).
2. Write the symbols for each element
3. Find the valency for each ion by using the Periodic Table, table of polyatomic ions or Roman numerals in the name of the compound.
4. Use the cross-over method to determine how many of each type of ion is needed to make the total charge zero.
5. Write the formula using subscripts (numbers at the bottom of symbols).

Note: For two ions of valency 2 e.g. $Mg^{2}+$ and $O^{2}-$, only one ion of each type is needed and the formula is MgO. Do not write 1 as a subscript – the absence of a subscript means that there is only one of this ion.

6. If there are two or more polyatomic ions in the formula, then the whole ion is enclosed in brackets and the subscript is outside the brackets. e.g. magnesium nitrate is $Mg(NO_3)_2$

Now work through the questions. Check your answers after each set.

Set 1: What ion does each of the following elements form?

a) barium

b) lithium

c) oxygen

d) aluminium

e) potassium

f) chlorine

g) sulfur

h) fluorine

i) calcium

j) phosphorus.

Set 2: Write the formula for each of these compounds below:

a) barium chloride

b) lithium sulfide

c) calcium oxide

d) sodium fluoride

e) aluminium nitride

f) magnesium iodide

g) potassium phosphide

h) aluminium sulphide

i) calcium carbide

j) magnesium bromide.

Set 3: Write the names of these compounds below:

a) Li_2CO_3

b) $NaNO_3$

c) $Mg(OH)_2$

d) $CaSO_4$

e) NH_4NO_3

f) K_3PO_4

g) $Al_2(SO_4)_3$

h) $(NH_4)_2CO_3$

i) NaOH

j) $BaCO_3$

Set 4: Complete the table

NOTE: When naming compounds of a transition element that may have multiple valencies. The valance of the metal in the compound must be shown by Roman numerals. e.g., $Ti(NO_3)_2$ is titanium(II)nitrate.

Name	Chemical formula
sodium sulfide	
calcium fluoride	
nickel (II) carbonate	
aluminium hydride	
hydrogen sulfate	
ammonium sulfate	
barium phosphate	
gold (I) oxide	
iron (III) sulfide	
lead (II) nitrate	
cobalt (II) hydroxide	
	AgI
	$CuCl$
	Li_2CO_3
	ZnH_2
	Mg_3P_2
	$Fe(OH)_2$
	Na_2SO_4
	$CrCl_3$
	K_3N